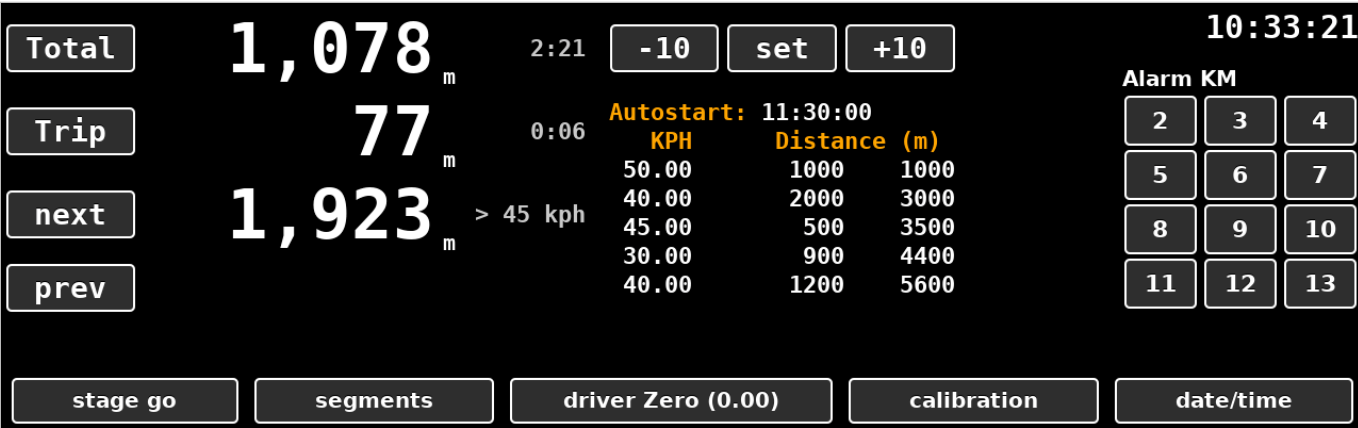
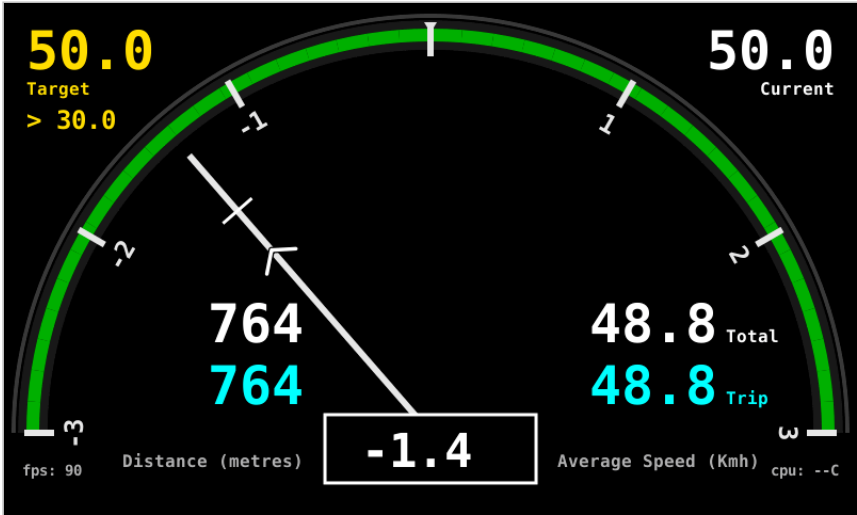


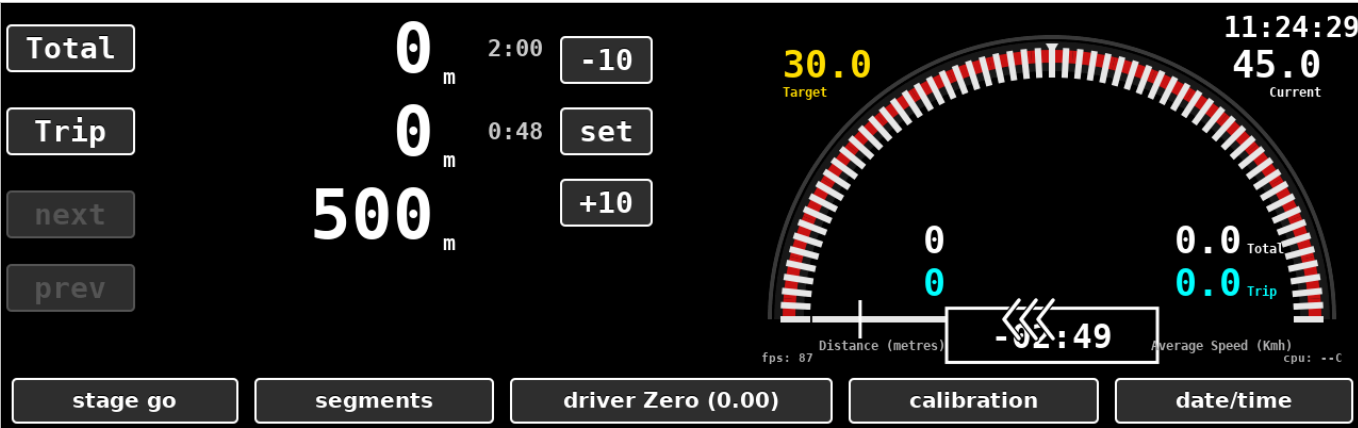
Rallybox User Manual



Co-pilot screen, two-display mode.



Driver display.



Co-pilot screen, single-display mode.

THE CO-PILOT MAIN SCREEN

Autostart: 11:30:00	
KPH	Distance (m)
50.00	1000
40.00	2000
45.00	500
30.00	900
40.00	1200

Total

Total displays distance and elapsed time since start. Pressing Total resets the distance to zero.

- Awaiting an autostart it also resets Trip but does not reset the timer.
- During a stage it also resets Trip but does not reset the timer. Confirmation is required.
- Outside of a stage it resets the timer but not the Trip.

Trip

Trip displays an intermediate distance and time since it was last pressed. If a stage is active it also resets itself with auto speed changes or when next or prev are pressed. During a stage or awaiting an autostart Trip will also be reset by Total.

next

next is pressed during a stage to move to next speed where the auto speed change check box has not been ticked, or to override the system if the speed change appears earlier than programmed. The following change point does not move and removes its same distance from the start.

The counter shows the distance still to run before the speed change and the speed it changes to. > END signifies there are no further speed changes, and ---. --- means no stage is loaded.

prev

prev undoes the last speed change, whether the crew made it or the box made it automatically; it works only in the section after the change and only once, and is grey when there is nothing to undo.

-10, set and +10

These correct the measured distance. -10 and +10 move Total and Trip by ten metres each press. set changes Total to the figure typed into the box (Trip is not affected). Changes made during a stage reestimate the average speeds and timing error.

Alarm KM

Pressing one of the numbers 2 to 13 sets an alarm when the car has travelled that many kilometres. A countdown of the distance appears under the pad, and clear cancels it.

The stage table

Shows each leg of the loaded stage with its target speed, its length, and the running total at its end.

DATE/TIME

DATE/TIME SETUP

System Clock: 2026/09/16 06:48:58

Rally Clock: 2026/09/16 06:38:11.1

Set Rally Clk: 2026/09/16 06:38:09

Options:

speed units **KPH**

force single display mode ☐

172.17.0.2:8080/

Save new rally clock entry back

The date/time screen.

Setting Rally Clock

The System Clock is the Raspberry Pi's own clock. The Rally Clock is the one every stage is timed against. Set Rally Clk is used to update the Rally Clock and is prepopulated ready for editing.

1. Tap the date box and delete what is there with <-.
2. Type the date on the keypad as yyyy/mm/dd, using the / key for the separators.
3. Tap the time box, delete what is there, and type the time as hh:mm:ss, using the : key.
4. Press Save new rally clock entry.

IMPORTANT: Both date and time boxes must be complete and formatted correctly otherwise the changes will not be saved. Do not enter tenths of seconds.

Trimming the Rally Clock

Small adjustments to the rally clock may be made by pressing the -0.1 and +0.1 buttons to adjust the time by a tenth of a second. Hold a button down to repeat. Each press is saved immediately.

Speed units

The button shows the unit in use. Pressing toggles between KPH and MPH and the selection is saved automatically.

Force Single Display Mode

When set to ON the driver's gauge will be incorporated into the co-driver screen when the program is next rebooted. The driver's display is disabled.

Phone Control Link

When the rally box is tethered to a mobile phone (prior to starting this program), the phone can be used to control the device and play the timing tones. The IP address shown on the device can be typed directly into the phone's browser or entered using the camera and QR code. Part 2 describes what the phone can do.

Exit App

exit app saves the settings and closes the program. It does not switch the Raspberry Pi off.

CALIBRATION

The screenshot shows the CALIBRATION screen with a black background and white text. At the top right is a timer showing 06:43:06. The main display area shows: 'Device distance: 0m Pulses 0 S1=0 S2=0', 'Actual distance covered: 500-100... metres' (where 500-100... is in a grey box), 'Using Sensors 1+2 (avg)' (where Sensors 1+2 (avg) is in yellow), and 'Using Calibration 9999 pulses/KM. Reset to []' (where 9999 pulses/KM is in yellow and [] is a grey box). Below this are two buttons: 'Set sensor 1' and 'Set both sensors and avg.'. At the bottom left are three buttons: 'Start/Zero', 'Save Calibration', and 'Back'. On the right side is a keypad with buttons for digits 0-9, a decimal point, a backspace key, and a 'C' key. At the bottom right, it says 'Prop/Wheel RPM at 8 pulses' and '0 rpm' (where 0 is in yellow).

CALIBRATION 06:43:06

Device distance: 0m Pulses 0 S1=0 S2=0

Actual distance covered: 500-100... metres

Using Sensors 1+2 (avg) Set sensor 1 Set both sensors and avg.

Using Calibration 9999 pulses/KM. Reset to [] Reset

INSTRUCTIONS:

1. Press Start/Zero to reset counter to zero or skip to step 2 to use existing counters.
2. Enter actual distance covered.
3. Press Sensor button to select counter.
4. Save Calibration

Prop/Wheel RPM at 8 pulses 0 rpm

Start/Zero Save Calibration Back

The calibration screen.

Calibrating over a set distance

1. Press Start/Zero to zero the counters.
2. Drive the measured kilometre or any known distance.
3. Enter the actual distance driven (not the device distance) into the Actual distance covered box. This must be at least 500 m.
4. Press Set sensor 1 or Set both sensors and avg. to choose which counter is used. This will remain displayed in yellow.
5. Press Save Calibration.

The saved calibration will be displayed in yellow, for example Calibration 9999 pulses/KM.

Adjusting an existing calibration

Where the calibration is incorrect it can be adjusted directly by entering a new pulses per kilometre figure into the Reset to box and pressing Reset. No driving is needed. A figure of zero or less is ignored. It is not necessary to press Save Calibration.

Prop/Wheel RPM

The reading under the keypad is the propshaft or wheel speed in revolutions per minute, taken from sensor 1. For it to be correct, type the number of pulses the sensor gives per revolution into the box ahead of the rpm reading.

Back

Back returns to the main screen.

SEGMENTS

The screenshot shows the 'STAGE SETUP' screen. It features a table with columns: Speed (KPH), Distance (m), Auto, and Time. The table contains five rows of data. To the right of the table are 'Set' and 'Recall' buttons numbered 1 to 5, and a 'clear memory' button. Further right are controls for 'Beep Assist KM' (0.25), 'Off/On' (toggle), 'Mode' (toggle), 'Advance by' (100), 'Navigation' (checked), 'Timing' (checked), and '0'. At the bottom right is a numeric keypad (0-9, *, /, C, <-) and a 'back' button. At the bottom left are 'New:' (KPH), 'metres (; sep...' (metres (; sep...), 'Auto' (checked), and 'add' buttons.

Speed (KPH)	Distance (m)	Auto	Time
50.00	1000	☒	01:12
40.00	2000	☒	03:00
45.00	500	☒	00:40
30.00	900	☒	01:47
40.00	1200	☒	01:48

The stage setup screen.

Entering new stage information

If there are existing speeds and distances shown, press `del` beside each line first.

1. Tap the KPH box and type the target speed.
2. Tap the metres box and type the length of the leg.
3. Tick Auto for the box to change to the next speed by itself at the end of the leg. Leave it clear to change speed by pressing next.
4. Press add and the entered information will populate the table.

Several legs can be entered at once by separating the figures with semicolons, e.g. `30;40` with `1000;2000` adds 1000 m at 30, then 2000 m at 40.

The table shows each leg with the time it should take. In a dual display configuration the total distance as well as the interim distances of the current loaded stage will show in the co-driver main window.

Saving and Recalling saved stages

Pressing Set 1 to 5 will save the stage on screen into that memory for later use, and you will be asked to confirm that you want to overwrite the memory slot if it is in use. The corresponding Recall button will invert colours.

Recall 1 to 5 loads it back into current memory and displays the details.

Clear Memory

Pressing this will clear all saved stages after confirmation. It does not delete the current stage information, which stays on the screen.

Tone

The driver's aural error tones can be enabled or disabled using the tone off/on switch. Type 1 pulses faster the larger the correction needed. Type 2 is a continuous tone whose pitch follows the seconds of error.

Beep Assist KM

Beep Assist is an aural navigation assistant which gives warnings ahead of upcoming navigations and alerts to identify measured distance inaccuracies.

The total roadbook elapsed distance at each box in a stage should be entered into the top entry field (not the distance between each box) and this will be saved and recalled alongside the stage speeds and distance information.

- Navigation assistance is designed to prevent a navigator missing an action by sounding an alert at a preconfigured distance, in metres, ahead of a roadbook box. 100 metres is a sensible advance configuration choice. Navigation gives a double beep.
- Timing assistance is designed to identify deviation away from perfect timings due to distance inaccuracies caused by wheel spin, corner cutting or sensor skipping. The alert sounds when the car should be passing a roadbook box. Zero seconds is a sensible Advance configuration choice, though it can be configured to beep early. Timing gives a single beep.

Beep Assist can be disabled using the Off/On switch. If it is enabled, the respective Navigation and Timing tick boxes need to be checked for it to sound. The advance configurations (metres and seconds) do not get saved and recalled.

back

back returns to the main screen.

STAGE GO

The screenshot shows a digital interface for a stage timer. At the top, there's a 'Total' button, a large '0' with a subscript 'm', a timer '0:51', and buttons for '-10', 'set', and '+10'. The top right corner displays the time '10:26:22'. Below the top bar, on the left, are buttons for 'Trip', 'next', and 'prev'. In the center, a large box titled 'Stage Go?' contains the text 'Stage = 1000m@50.00kph, 2000m@40.00kph' and 'Total and Trip Counters will reset'. Below this text are five buttons: 'Now', 'Later', 'At 10:26', 'ABORT STAGE', and 'Back'. To the right of the central box is a numeric keypad with buttons 3, 4, 6, 7, 9, 10, 12, and 13. Above the keypad is a label 'Alarm KM'. At the bottom of the screen, there are five buttons: 'stage go', 'segments', 'driver Zero (0.00)', 'calibration', and 'date/time'.

The question the stage go button asks.

Stage Go displays the loaded stage details for review and offers three ways to start.

- Now starts the stage immediately.
- At HH:MM sets an autostart at the next round minute and permits early departure.
- Later up to 3 hours in the future. Can be set while a stage is in progress. No early departure.

Now

Starts the stage immediately and has the following characteristics:

- locks the current stage data for the remainder of the stage.
- cancels any stage running and other future programmed autostarts.
- resets Total and Trip distances to zero and the counter starts the timer immediately.

Advantages: can be used with a flying start if late to a stage, though the difference between the start time and the actual time at the start line will need to be corrected for. Disadvantages: start time accuracy is only as good as the navigator's reactions.

At HH:MM

Starts the stage on the next whole minute, which is automatically shown on the button. It has the following characteristics:

- locks the current stage data for the upcoming stage.
- cancels any stage running and other future programmed autostarts.
- displays stage speed and countdown to the start in the driver's display.
- early departure is enabled, allowing the car to pull away early if the car will struggle to get up to speed or will be obstructed. This works because the timer resets at the autostart minute but the distance is reset when the option is selected.
- pressing Total ahead of departure will zero any pre-departure movements.
- if after the HH:MM menu option appears you wait until after the minute has passed, you will be told you need to trigger the menu again as the start time elapsed.

Advantages: can make an early departure; simple one button press when lining up for the start; perfect start time.

Disadvantages: can only be pressed within 59 seconds of the start.

Later

Starts the stage a predetermined time up to 3 hours in the future, even if a stage is currently under way. To use it tap the autostart entry box, type the start time as hh:mm:ss on the keypad, press set to store; back to return to main screen. Press Clear to remove the autostart. It has the following characteristics:

- it does NOT lock the current stage data for the upcoming stage. Whatever stage data is loaded when the autostart fires will only be locked at that time.
- it does NOT cancel any stage running. You can continue with the existing stage.
- it displays a countdown, but the driver display will continue to display the current stage information.
- early departure is NOT enabled, because the distances and timer reset at the autostart time.

Advantages: can be configured while another stage is in progress; can be configured more than 1 minute in advance; can make last minute changes to segment instructions; perfect start time. Disadvantages: no early start; very early reconfiguration means the driver display is obscured by the countdown.

ABORT STAGE

Ends the stage after confirmation. The driver's needle returns to zero and the tone stops. Total and Trip keep their readings, and any autostart that has been set is cancelled.

Back

Closes the question and changes nothing.

DRIVER ZERO

This option is used to rebase the driver's gauge to show zero error when the driver targets an error other than zero.

The popup window will show the ahead/behind value (the true error) and the current adjustment (how much is currently adjusted).

- Yes sets the offset to zero.
- Reset to 0.0 removes the offset. The gauge shows the true error again.
- No changes nothing.

Only the driver's gauge changes. The distances, the schedule and the clock stay as they are, and starting a stage clears the offset.

Total314m0:32-10set+10

Trip3168

nextprev

Adjust the ahead/behind value
by -10.02 seconds,
currently 0.00 seconds

YesNoReset to 0.0

Alarm KM

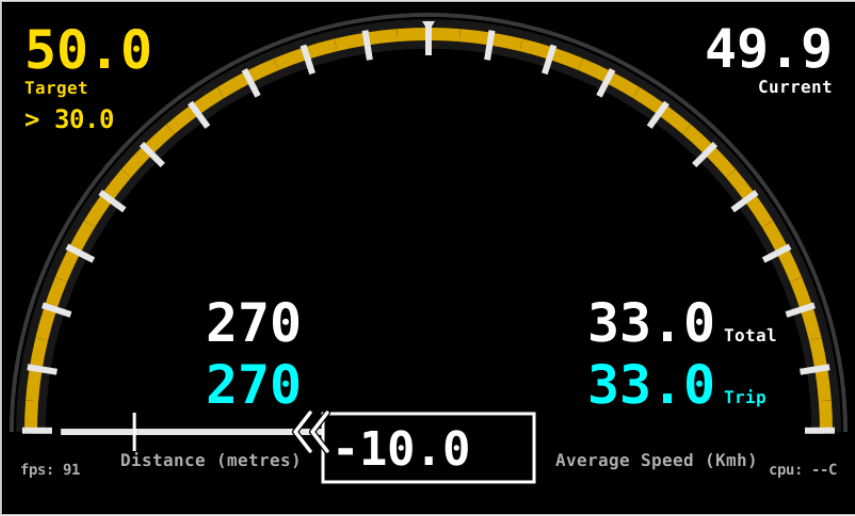
234

567

8910

111213

stage gosegmentsdriver Zero (0.00)calibrationdate/time



THE PHONE APP

Connecting: The rally box must first be tethered to the phone using the box's wifi connector. This web app can then be connected by typing the IP address shown on the date/time page of the box into the browser or via the QR code. The phone page will show connected in green when active.

Sounds: The top of the phone screen (see below) will display Tap For Sound in orange if the link is inactive. Otherwise it toggles between Sound is On and Sound is Off.

On **Live** (co-pilot Main) and **Driver** (driver/main gauge) press the Total or Trip box to reset distance; Live also allows prev/next under the rally clock. **Setup** replicates Segments.

Live: replicates Main.

Sound is onconnected

Live

Setup

Driver

Rally Clock

09:34:39

prev

next

+1.1 s

Total

915 m

avg 35.1 kph 34.9

Trip

414 m

avg 31.0 tgt 30.0

Adjust

-10

Total m

set

+10

Segment

2

of

3

Autostart: none

KPH

Distance (m)

40.00

500

500

30.00

500

1,000

35.00

500

1,500

Driver: replicates Driver/Main.

Tap for soundaudio: engine stalledconnected

Live

Setup

Driver

09:37:04

35.0

Target

35.1

Current

+2.5

Total

2,322 m

avg 35.1

Trip

1,322 m

avg 35.0

Adjust

-10

Total m

set

+10

Setup: replicates Segments.

Sound is onconnected

Live

Setup

Driver

#	Target kph	Distance m	Auto	
1	40	500	☒	del
2	30	500	☒	del
3	35	500	☒	del

+ Add segment

Store

1

2

3

4

5

Recall

1

2

3

4

5

Beep Assist KM

0.50, 1.00

Off/On

☒

Mode

Advance by

Navigation

☒

100

Timing

☒

0

tone off/on

☒

Type

☐ 1

☒ 2

Reset Trip

Reset Total